Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E.V, Computer Science and Engineering
Course Code & Title: CCS339 Cryptocurrency and Blockchain technologies
Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit I, IV, V / Cryptographic Hash Functions, Solidity, Case study

Course Outcome: CO 1, CO 4, CO 5

Unit Learning Outcome: UO 1c, 4a, 5a

Activity Chosen: Demonstration and Practice

Justification:

Hands-on training is one method used by educational systems to teach people how to perform a specific task. It gives the student real-world experience by allowing him or her to get his or her hands directly on whatever he or she is learning. It provides a clear example of how something is done or how a concept works. Students are forced to confront their prior understanding of a concept as a result of this activity.

Time Allotted for the Activity: 20 minutes

Details of the Implementation:

<u>Blockchain Demo</u>

- Students were told to open any web browser and navigate to https://andersbrownworth.com/blockchain/.
- Make the students practice on the blockchain demo website to learn more about the concepts in depth by practicing.
- Students experienced the block creation and hash working concepts visually.
- Students shared their experiences with their peer students.

<u>RemixIDE & Truffle</u>

- Students were told to open any web browser and navigate to the official RemixIDE website and Truffle.
- The default environment is the "Solidity" environment for smart contract development.
- Open a new file and name it with the extension. sol.
- Students experienced different datatypes in the Solidity programming language and also practiced Hello World and addition.
- Students were given knowledge on how to write smart contracts and the deployment process.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PSO1	PSO3
CO6	**2**	**2**	**2**	**2**	**3**	**2**	**2**

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PSO2**	**PSO3**
	2	**2**	**2**	**2**	**3**	**2**	**2**
Justification for correlation	Students will be able to understand the concept of blockchain technology with Engineering Knowledge	Students will analyze blockchain use cases and scenarios using engineering sciences.	Students will design and develop blockchain-based solutions by implementing smart contracts and decentralized applications.	Students will investigate complex problems related to blockchain models and explore potential improvement	Students will effectively use blockchain development tools and platforms to implement and gain hands-on experience in blockchain technology.	Students will be able to apply blockchain concepts to develop reliable IT solutions.	Students will be able to incorporate blockchain with AI to provide solutions to real-world problems in Industry.

Screenshot of the practice:

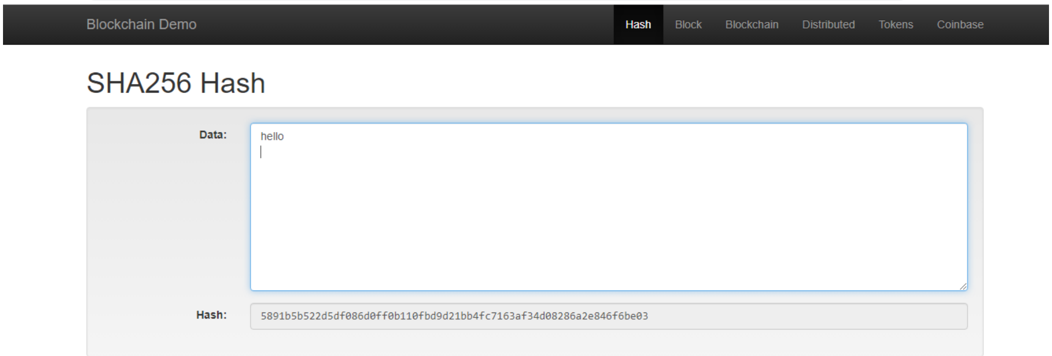

Fig:1 Demonstration of SHA256 Hash

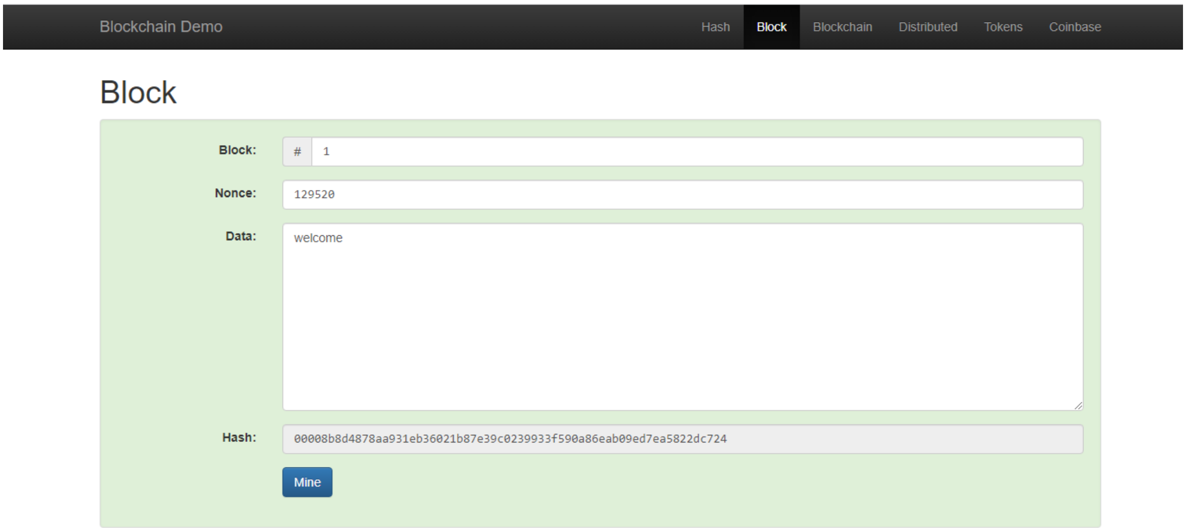

Fig:2 Demonstration of block in a blockchain

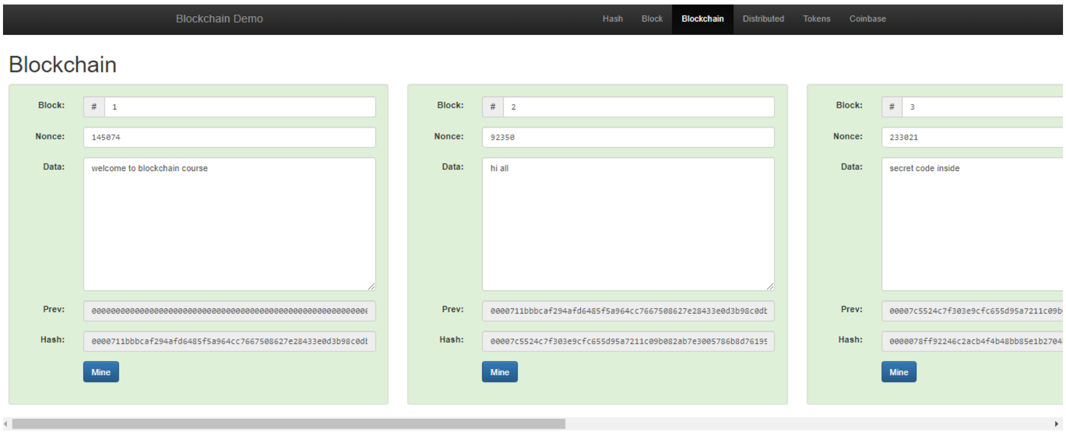

Fig:3 Demonstration of blockchain

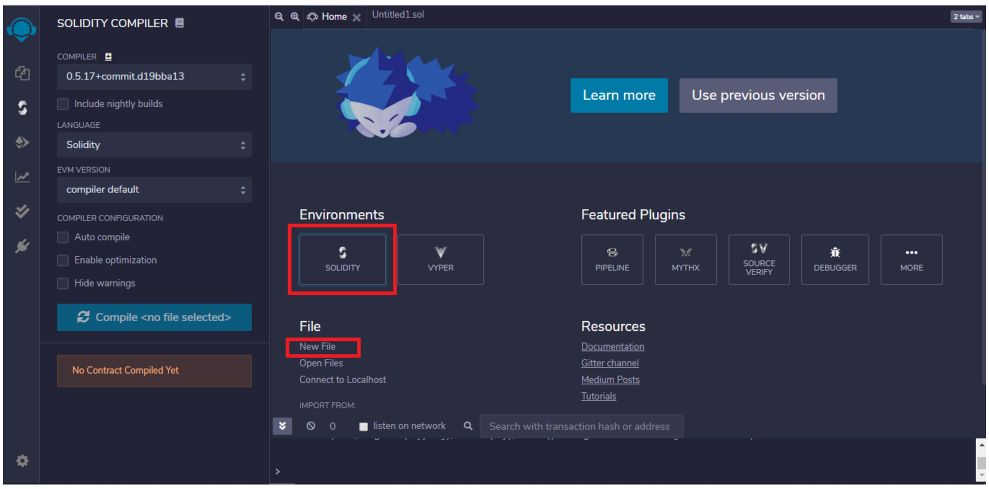

Fig:4 Demonstration of RemixIDE

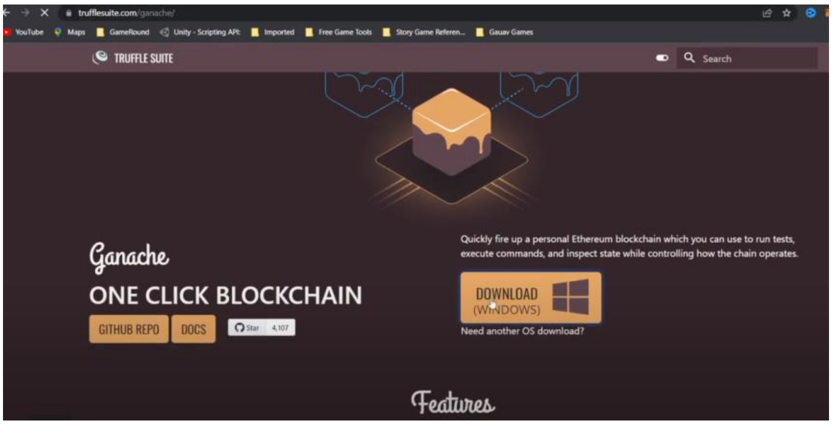

Fig:5 Demonstration of Trufflesuite

Reflective Critique:

- Students know the possible online tools related to the subjects.
- The demo activity increases the student's presentation skills, knowledge and communication skills.
- Students get in-sight knowledge about the particular topics.

Benefit of the practice:

- The students enjoyed the activity.
- Hands-on practice aids in depth about concept clearly.
- Through practice, students develop the ability to apply their knowledge and skills to real-world challenges.

Challenges faced in implementation:

- Remix IDE is designed as an online environment, and its integration with local development environments is limited. So, not able to deploy in local environment.

References:

1. https://andersbrownworth.com/blockchain/
2. https://remix.ethereum.org/#lang=en&optimize=false&runs=200&evmVersion=null&version=soljson-v0.8.22+commit.4fc1097e.js
3. https://www.teacheracademy.eu/blog/online-apps-tools-for-teaching/
4. https://www.european-agency.org/sites/default/files/itlresearh2011findings.pdf

Signature of Faculty Member **HOD**